# A Coifman–Rochberg-Type Characterization of Dyadic John–Nirenberg Space

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**Abstract**

Kinnunen and Myyryläinen ask whether dyadic John–Nirenberg space admits a Coifman–Rochberg-type characterization. We prove that, on every finite cube and for $1 < p < \infty$, dyadic $JN_p$ is exactly the difference of its positive dyadic $A_1$ cone. The decomposition is quantitative and its $A_1$ constants are uniform. Each positive summand also has an exact representation as a bounded positive factor times a fractional power of a dyadic maximal function. The proof is a Rubio de Francia iteration made possible by the maximal-operator theorem in the source paper and the weak-$L^p$ embedding of $JN_p$.

## 1 The question and the answer

Let $Q_0 \subset \mathbb{R}^n$ be a finite cube, let $\mathcal{D}(Q_0)$ be its dyadic subcubes, and let $1 < p < \infty$. Write

$$[f]_{JN_p^{\mathrm{d}}(Q_0)} := \sup_{\mathcal{F}} \left( \sum_{Q \in \mathcal{F}} |Q| \left( \fint_Q |f - f_Q| \right)^p \right)^{1/p},$$

where the supremum is over pairwise disjoint families $\mathcal{F} \subset \mathcal{D}(Q_0)$. Constants are the kernel of this seminorm. We use the normalized Banach norm

$$\|f\|_X := |Q_0|^{1/p} |f_{Q_0}| + [f]_{JN_p^{\mathrm{d}}(Q_0)}. \tag{1}$$

For a nonnegative locally integrable $w$, write $w \in A_1^{\mathrm{d}}(Q_0)$ when

$$M_{Q_0}^{\mathrm{d}} w \le [w]_{A_1^{\mathrm{d}}(Q_0)} w \qquad \text{a.e. on } Q_0.$$

On page 2 of arXiv:2107.00492v2, the source paper asks whether a Coifman–Rochberg-type characterization exists for dyadic John–Nirenberg space. It proves that $M_{Q_0}^{\mathrm{d}}$ is bounded on the $JN_p^{\mathrm{d}}$ seminorm and, motivated by the inclusion $(M_{Q_0}^{\mathrm{d}} h)^{1/p} \in JN_p^{\mathrm{d}}$ for $h \in L^1$, asks whether a Coifman–Rochberg-type characterization exists. The following gives a precise affirmative answer.

**Theorem 1** (Dyadic $A_1$ difference characterization). *For every finite cube $Q_0$ and $1 < p < \infty$,*

$$JN_p^{\mathrm{d}}(Q_0) = (JN_p^{\mathrm{d}}(Q_0) \cap A_1^{\mathrm{d}}(Q_0)) - (JN_p^{\mathrm{d}}(Q_0) \cap A_1^{\mathrm{d}}(Q_0)). \tag{2}$$

*More quantitatively, there is $C = C(n,p)$ such that every real $f \in JN_p^{\mathrm{d}}(Q_0)$ can be written $f = u - v$, where $u, v > 0$ a.e.,*

$$u, v \in JN_p^{\mathrm{d}}(Q_0) \cap A_1^{\mathrm{d}}(Q_0), \qquad [u]_{A_1^{\mathrm{d}}(Q_0)} + [v]_{A_1^{\mathrm{d}}(Q_0)} \le C, \qquad \|u\|_X + \|v\|_X \le C\|f\|_X.$$

*Conversely, every such difference belongs to $JN_p^{\mathrm{d}}(Q_0)$, and*

$$\|f\|_X \le \|u\|_X + \|v\|_X.$$

*There are also an exponent $r = r(n,p) > 1$, functions $h_+, h_- \in L^1_+(Q_0)$, and measurable multipliers $k_+, k_-$ satisfying $C^{-1} \le k_\pm \le 1$ such that*

$$f = k_+(M^{\mathrm{d}}_{Q_0} h_+)^{1/r} - k_-(M^{\mathrm{d}}_{Q_0} h_-)^{1/r}, \tag{3}$$

*and the two displayed summands are precisely $u$ and $v$ above.*

## 2 Idea of the proof

The source theorem controls the oscillation seminorm of $M^{\mathrm{d}}_{Q_0} f$, but the Rubio de Francia algorithm needs a genuine Banach norm. The weak-$L^p$ embedding of $JN_p$ supplies the missing average estimate, so $M^{\mathrm{d}}_{Q_0}$ is bounded for (1). Absolute value is also bounded for this norm.

Starting from $g = |f|$, iterate the dyadic maximal operator:

$$\mathcal{R}g = \sum_{j\ge 0} \frac{(M^{\mathrm{d}}_{Q_0})^j g}{(2A)^j}, \qquad A = \|M^{\mathrm{d}}_{Q_0}\|_{X\to X}.$$

Then $\mathcal{R}g \ge g$, its $X$ norm is controlled, and $M^{\mathrm{d}}_{Q_0}(\mathcal{R}g) \le 2A\mathcal{R}g$. Thus $\mathcal{R}|f|$ is an $A_1$ majorant of $|f|$. The two functions $2\mathcal{R}|f| + f$ and $2\mathcal{R}|f|$ are positive, remain in $JN_p$, and are comparable to that majorant, hence are uniformly $A_1$. Finally, reverse Hölder self-improvement of $A_1$ weights converts each summand to the maximal-power form (3).

## 3 The two norm estimates

**Lemma 2** (Absolute value)**.** *There is an absolute constant $C$ such that $\||f|\|_X \le C\|f\|_X$.*

*Proof.* For every dyadic $Q$ and every constant $c$,

$$\fint_Q \big||f| - (|f|)_Q\big| \le 2\fint_Q \big||f| - |c|\big|.$$

Taking $c = f_Q$ gives $[|f|]_{JN_p^{\mathrm{d}}(Q_0)} \le 2[f]_{JN_p^{\mathrm{d}}(Q_0)}$. On $Q_0$,

$$|Q_0|^{1/p}(|f|)_{Q_0} \le |Q_0|^{1/p}|f_{Q_0}| + |Q_0|^{1/p}\fint_{Q_0} |f - f_{Q_0}| \le |Q_0|^{1/p}|f_{Q_0}| + [f]_{JN_p^{\mathrm{d}}(Q_0)}.$$

Combining the two bounds proves the claim. □

**Lemma 3** (Full-norm maximal estimate)**.** *The local dyadic maximal operator $M^{\mathrm{d}}_{Q_0}$ is bounded on $(JN_p^{\mathrm{d}}(Q_0), \|\cdot\|_X)$.*

*Proof.* Kinnunen–Myyryläinen, Theorem 4.2, gives

$$[M^{\mathrm{d}}_{Q_0} f]_{JN_p^{\mathrm{d}}(Q_0)} \le C_{n,p}[f]_{JN_p^{\mathrm{d}}(Q_0)}. \tag{4}$$

The dyadic John–Nirenberg inequality gives the standard embedding

$$\|f - f_{Q_0}\|_{L^{p,\infty}(Q_0)} \le C_{n,p}[f]_{JN_p^{\mathrm{d}}(Q_0)}. \tag{5}$$

Fix $1 < q < p$. The finite-measure Lorentz embedding and the strong $(q, q)$ maximal inequality imply

$$\begin{aligned}\fint_{Q_0} M^{\mathrm{d}}_{Q_0}(f - f_{Q_0}) &\le |Q_0|^{-1/q}\|M^{\mathrm{d}}_{Q_0}(f - f_{Q_0})\|_{L^q(Q_0)} \\ &\le C_{n,q}|Q_0|^{-1/q}\|f - f_{Q_0}\|_{L^q(Q_0)} \le C_{n,p}|Q_0|^{-1/p}[f]_{JN_p^{\mathrm{d}}(Q_0)}.\end{aligned}$$

Since $M^{\mathrm{d}}_{Q_0} f \le M^{\mathrm{d}}_{Q_0}(f - f_{Q_0}) + |f_{Q_0}|$, this controls the average term in (1). Together with (4), it proves the lemma. Completeness follows from completeness modulo constants proved in the source paper plus the scalar mean coordinate. □

## 4 Rubio de Francia iteration and the decomposition

*Proof of Theorem 1.* Let $A \ge 1$ be the operator norm from Lemma 3. For $g \ge 0$ in $X$, define

$$\mathcal{R}g := \sum_{j=0}^{\infty} \frac{(M^{\mathrm{d}}_{Q_0})^j g}{(2A)^j}. \tag{6}$$

The series converges in $X$, because its $j$th term has norm at most $2^{-j}\|g\|_X$. The embedding $X \hookrightarrow L^1(Q_0)$ and monotone convergence identify this norm limit with the pointwise nonnegative series. Consequently

$$g \le \mathcal{R}g, \qquad \|\mathcal{R}g\|_X \le 2\|g\|_X, \qquad M^{\mathrm{d}}_{Q_0}(\mathcal{R}g) \le 2A\mathcal{R}g. \tag{7}$$

The last inequality follows from sublinearity, first for partial sums and then by monotone convergence. Thus every nonzero $g$ has a positive dyadic $A_1$ majorant with uniformly controlled constant.

For $f \ne 0$, apply this to $g = |f|$ and put $w = \mathcal{R}|f|$. Set

$$u = 2w + f, \qquad v = 2w.$$

Then $f = u - v$ and, because $w \ge |f|$,

$$w \le u \le 3w, \qquad v = 2w.$$

Hence $u, v > 0$ a.e. and

$$M^{\mathrm{d}}_{Q_0} u \le 3M^{\mathrm{d}}_{Q_0} w \le 6Aw \le 6Au, \qquad M^{\mathrm{d}}_{Q_0} v \le 2Av.$$

Thus $u, v \in A_1^{\mathrm{d}}(Q_0)$. They lie in $JN_p^{\mathrm{d}}(Q_0)$ by linearity, and Lemma 2 together with (7) gives the claimed norm bound. If $f = 0$, take $u = v = 1$. The reverse inclusion in (2) and its norm inequality follow immediately from linearity and the triangle inequality.

It remains to prove the maximal-power refinement. The $A_1$ constants of $u, v$ are bounded solely in terms of $n, p$. The dyadic reverse Hölder theorem therefore supplies one uniform $r > 1$ and $B < \infty$ such that, for $z = u, v$ and every $Q \in \mathcal{D}(Q_0)$,

$$\left(\fint_Q z^r\right)^{1/r} \le B \fint_Q z. \tag{8}$$

In particular $h = z^r \in L^1(Q_0)$. Dyadic differentiation and (8) yield, almost everywhere,

$$z \le (M^{\mathrm{d}}_{Q_0} h)^{1/r} \le B M^{\mathrm{d}}_{Q_0} z \le B[z]_{A_1^{\mathrm{d}}(Q_0)} z.$$

Therefore

$$z = k(M^{\mathrm{d}}_{Q_0}(z^r))^{1/r}, \qquad k := \frac{z}{(M^{\mathrm{d}}_{Q_0}(z^r))^{1/r}}, \qquad (B[z]_{A_1^{\mathrm{d}}(Q_0)})^{-1} \le k \le 1.$$

Apply this to $u$ and $v$ to obtain (3). □

# 5 Scope and novelty check

The theorem is an exact difference-cone characterization and the refinement is an exact maximal-power formula with bounded positive multipliers. It does not assert the stronger identity obtained by requiring $k_+ = k_- = 1$ and the fixed exponent $1/p$. The source question does not prescribe that stronger form.

Focused searches through 13 August 2026 found later work on dyadic $JN_p$ preduals, vanishing subspaces, and fractional dyadic maximal operators, but no statement of (2) or (3). The proof is also consistent with the classical Coifman–Rochberg philosophy: maximal boundedness produces an $A_1$ cone, and the ambient oscillation space is its difference hull.

# References


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